

Connecting to Compete

2025

Trade Logistics in the Global Economy



The New Logistics Performance Indicators 2.0

Connecting to Compete 2025

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Abbreviations

B2B	Business-to-business
B2C	Business-to-consumer
LLDC	Landlocked developing country
LPI	Logistics Performance Index (used through 2023)
LPI 2.0	Logistics Performance Indicators 2.0 (used from 2025 onward)
PPP	public–private partnership
SMEs	small and medium enterprises

About this report

Connecting to Compete 2025 introduces the Logistics Performance Indicators 2.0 (LPI 2.0), a fundamental redesign of the World Bank's former survey-based Logistics Performance Index toward a system grounded in shipment-level data. Drawing on large-scale tracking data from maritime, aviation, and postal operators—together accounting for over 80 percent of global goods trade—the LPI 2.0 provide standardized and comparable measures of supply chain connectivity, speed, and reliability across economies and over time. The framework comprises 21 indicators, including 6 core indicators capturing direct connectivity and import time performance in each mode and 15 supplementary indicators that shed light on maritime

competition, transshipment dependence, corridor performance for landlocked developing countries, and postal logistics for e-commerce. Data for 2023–24 reveal large variations in connectivity across regions, substantial time penalties associated with transshipment and border procedures, and high unpredictability concentrated at ports, transshipment hubs, and inland checkpoints—particularly for imports and for landlocked countries. By shifting the focus from perceptions to observed outcomes, the LPI 2.0 significantly enhance diagnostic precision and policy relevance, enabling governments to better prioritize reforms with the highest potential impact on supply chain connectivity, speed, reliability, and resilience.

Executive summary

Logistics underpins the movement of goods and the operation of supply chains within and across borders. Efficient logistics reduces trade costs, connects firms to markets, and supports development, greatly increasing the volume and value of trade. Recent disruptions have underscored that resilient, well-functioning global logistics networks are essential for growth and welfare. They help countries withstand shocks, such as natural disasters, conflict, and technical failures, by ensuring reliable access to strategic goods that are too costly or impossible to produce domestically.

The Logistics Performance Indicators 2.0 (LPI 2.0) represent a major methodological shift toward shipment-level, real-world evidence. Drawing on large volumes of tracking data from maritime, aviation, and postal operators—modes that jointly account for more than 80 percent of the global goods trade—the LPI 2.0 provide standardized indicators of logistics performance across economies and time. Unlike the original Logistics Performance Index (LPI), which was based on surveys of logistics professionals, the LPI 2.0 measure the speed, reliability, and connectivity of supply chains based on observed shipment data.¹

The LPI 2.0's 21 indicators provide deep insight into the functioning of logistics networks. Six core indicators (one connectivity indicator and one time indicator for maritime, air, and postal logistics) and 15 supplementary (connectivity and time) indicators provide insights into maritime competition, transshipment burdens, corridor performance for landlocked developing countries (LLDCs), and postal logistics for e-commerce. Core connectivity indicators capture the number of countries that can be reached directly via maritime, air cargo, or postal services, while core time indicators measure import

dwell times and delivery times—key proxies for logistics efficiency and reliability.

The shift to LPI 2.0 expands coverage, enhances comparability, and deepens diagnostic power, enabling policymakers to identify binding constraints with far greater precision. Data on all 21 indicators are published for 2023 and 2024 at <https://lpi.worldbank.org>, with options for comparisons across economies and over time.

Five main findings stand out from the analysis of the LPI 2.0:

- **There are large connectivity gaps across and within regions.** On average, the Europe and Central Asia region and the North America region tend to exhibit the strongest connectivity, depending on the logistics mode, while Sub-Saharan Africa and Latin America and the Caribbean lag behind. These patterns reflect income, geography, and market accessibility. Within regions, some developing economies, such as Malaysia, Morocco, and Sri Lanka, showcase stronger performance due to their emergence as regional hubs.
- **Direct connections matter.** Roughly half of containers worldwide require at least one transshipment. Each transshipment adds about 18 days to maritime lead times, creating structural disadvantages for small, remote economies.
- **Several low- and middle-income economies clear imported containers within a few days, which is comparable to high-income economies, while others experience import dwell times exceeding 10 days.** Longer dwell times are due to weak customs automation, inefficient risk management, and cumbersome inspection and clearance processes. Aviation dwell times are

shorter than those for containers because air cargo typically involves smaller volumes, higher value goods, and more integrated logistics systems.

- **LLDCs face some of the longest and most unpredictable import times globally, with most delays stemming from idling at ports, border controls, and inland checkpoints.** LLDCs must transit coastal countries, often crossing multiple borders and facing complex transit procedures. Corridor lead times and dwell times at the port of arrival dominate total import time, underscoring the importance of regional coordination, interoperable transit guarantees, and streamlined border procedures to improve reliability.
 - **Delays are most unpredictable when goods are stationary—in ports and transshipment facilities—and more so for imports than exports.** Aviation supply chains show especially high dispersion. Unpredictability imposes heavy costs on firms: it disrupts production planning, increases inventory needs, and reduces competitiveness in just-in-time and perishable goods markets. Digital visibility systems, such as those implemented in India and the United States, demonstrate how the value of real-time tracking can mitigate these risks.
- Although logistics services are provided mainly by private operators, public policy is fundamental to the performance of global supply and value chains. The LPI 2.0 enable governments to prioritize interventions with the highest potential impact on supply chain speed, reliability, and resilience, tailored to country circumstances. Policy interventions should be implemented in a targeted, coherent manner across three strategic areas:
- **Enhancing connectivity.** Open transport and logistics markets to competition, invest in multimodal corridors and logistics clusters, and use regional integration and shared connectivity solutions so smaller or remote economies can benefit from economies of scale.
 - **Reducing time and variability.** Modernize border management through digitalization and risk-based controls (trade facilitation), improve port and airport performance, and strengthen traceability to make supply chains faster and more predictable.
 - **Strengthening institutions and regulation.** Build a robust governance framework with transparent public–private partnerships, competitive logistics markets, and coordinated policies across ministries, departments, and agencies to sustain and scale logistics reforms.

Introducing the 2025 LPI 2.0

Why is logistics important for trade and development?

Logistics consists of a set of activities (freight transportation, warehousing, brokerage) that move goods and establish supply chains across and within borders. Logistics performance captures the efficiency of supply chain connections, measured in time, cost, and reliability. Since 2007 the Logistics Performance Index (LPI) initiative has emphasized the importance of efficient supply chains in connecting firms to markets, reducing trade costs, and supporting development.² Logistics performance supports economic development by enabling trade within and across borders.³

Economics literature, often using the survey-based LPI, has established that logistics performance positively and significantly impacts the volume and value of bilateral trade, in particular exports.⁴ Logistics influences the time required for a product to reach its market. On average, an increase in inland transit time of one day reduces exports by 7 percent.⁵ For manufactured goods exports the likelihood of a country exporting to the United States decreases by 1.5 percent for each additional day of shipping time.⁶

Logistics inefficiencies hamper economies' competitiveness because logistics affect manufacturing firms' production costs.⁷ Improved logistics performance reduces production costs and is linked to stronger participation in global value chains.⁸ This is achieved not just through lower direct logistics costs (transport, storage, and administrative costs) but also by reducing the need for inventory and working capital.⁹

A series of disruptions of global logistics networks in recent years has underlined how well-functioning logistics networks are essential to production and welfare worldwide.¹⁰

Resilient logistics networks strengthen economies' capacity to withstand natural disasters, conflict, technical failures, or other shocks by securing reliable access to strategic goods (such as pharmaceuticals, food, energy, and technologies) that are prohibitively costly to produce domestically or cannot be produced at all.

Conceptual framework of the LPI 2.0 and evolution from the original LPI

Until 2023 the LPI was based on a survey in which logistics professionals rated, on a 1–5 scale, the ease of establishing supply chains with economies they traded with. When the LPI was conceived in the mid-2000s, perception-based data were the only option. However, survey-based indicators have limitations, including perception bias and large year-on-year fluctuations.

The Logistics Performance Indicators 2.0 (LPI 2.0) move away from perceptions and rely instead on objective measures derived from large volumes of shipment-level tracking data made available to the World Bank on a confidential basis by global industry and international tracking initiatives. These data provide granular measures of the speed and connectivity of international supply chains worldwide. Drawing on observations of logistics activity collected throughout the year, the LPI 2.0 offer a more comprehensive view of structural performance and reduce the influence of temporary shocks or short-lived events on the results.

Three core components define logistics performance: time, reliability, and cost. The LPI 2.0 measure time and reliability and capture features of logistics services that influence costs. They do so through indicators in two areas: connectivity and time. Connectivity refers to the number of markets an economy can directly trade with without having to go through intermediaries

and the number and types of services available. Time refers to the time that logistics processes take and their dispersion, which captures the reliability of the logistics processes. Both connectivity and time are measured for different logistics modes. Cost is not directly measured by the LPI 2.0; however, two connectivity indicators highlight competition within the maritime sector, higher degrees of which tend to be associated with lower logistics costs.¹¹

The 2025 LPI 2.0 cover three logistics modes—maritime, aviation, and postal—capturing over 80 percent of global goods trade by value.¹² The digitization of global supply chains means that shipments are now tracked globally in real time. Though digitization has reached all transport modes and most logistics services, tracking data are available in a globally uniform format only for maritime, aviation, and postal logistics.¹³

The new indicators

The indicators that make up the LPI 2.0 measure the connectivity and time of international trade supply chains, including the time shipments spend at key nodes such as seaports and airports. The LPI 2.0 consist of 21 indicators, each reported at the economy level and with different statistical measures.¹⁴

Core indicators

There are six core indicators in the LPI 2.0, one connectivity indicator and one time indicator

for each of the three logistics modes covered (maritime, air, and postal) (table 1.1). The core maritime connectivity indicator is the number of unique economies that are reachable from and that can reach a given economy through direct maritime services operated by liner shipping companies on a predefined rotation.¹⁵ The core aviation connectivity indicator is the average number of economies that are reachable from and that can reach a given economy through direct cargo-carrying air transport services.¹⁶ The core postal connectivity indicator is the average number of economies with inbound and outbound postal connections for parcel and express shipments with a given economy.¹⁷ Parcel and express shipments are primarily business-to-business (B2B) transactions and are thus comparable to maritime container and air cargo shipments, which are purely B2B.

Connectivity is positively associated with logistics performance across all modes. More connected economies can directly reach a larger number of trade partners and tend to face lower trade costs. Higher connectivity increases competition among logistics service providers and encourages investment in trade-related infrastructure, such as ports and information technology systems. It is also associated with a stronger presence of global logistics operators, which facilitates the diffusion of best practices and can generate positive spillovers for domestic logistics services. Connectivity is also critical for resilience. Economies that rely on only one or a few international partners are more vulnerable to disruptions when those links break down,

Table 1.1 The six core indicators of the LPI 2.0		
Logistics mode	Connectivity indicators	Time indicators
Maritime	• <i>Number of maritime partner economies:</i> Number of economies accessible through direct liner shipping services.	• <i>Container import dwell time (days):</i> Time spent by containers at the port of entry from arrival (ship unloading) to departure from the port.
Aviation	• <i>Number of aviation partner economies:</i> Number of economies accessible through direct air cargo connections.	• <i>Aviation import dwell time (days):</i> Time spent by goods at the airport between the moment the cargo is ready to be picked up by the consignee or their agent from the airport and the moment the cargo is cleared by customs and leaves the airport.
Postal	• <i>Number of business-to-business postal partner economies:</i> Number of economies accessible through direct international postal connections for B2B shipments.	• <i>Business-to-business postal delivery time (days):</i> Time spent by a postal item (parcel or express shipment) in the importing economy from arrival at the economy's office of exchange (postal bureau) to the first attempted or final delivery.

Source: World Bank.

whether due to natural disasters, conflict, technical failures, or other shocks.

The core time indicators measure the time it takes to import cargo. The core maritime and aviation time indicators are container import dwell time and aviation import dwell time, respectively. The core postal time indicator is the business-to-business delivery time, which captures customs clearance time and the time until the first attempted or final delivery. Import dwell and delivery times were selected as core indicators because customs procedures are generally more complex and time-consuming for imports than for exports, except in economies with strict export controls. In addition, for air cargo and postal shipments, event tracking is typically more complete and consistent for imports than for exports.

Import dwell and delivery times capture the efficiency of several components of trade logistics, including customs clearance and

border management; seaport and airport operations (such as terminal operations, cargo handling, and yard management); cargo removal practices;¹⁸ land transport services (including the availability of trucks or other equipment for onward transport); and the quality of transport and logistics infrastructure.

Supplementary indicators

While the six core indicators provide a solid overview of an economy's logistics performance, the LPI 2.0 provide additional connectivity and time indicators for maritime and postal logistics to measure performance more precisely (table 1.2). Two supplementary maritime connectivity indicators—the number of direct container liner shipping services and alliances servicing the ports in an economy—highlight competition within the sector,¹⁹ higher degrees of which tend to be associated with better performance

Table 1.2 Supplementary indicators of the LPI 2.0

Logistics mode	Connectivity indicators	Time indicators
Maritime	• <i>Number of services</i>: Number of direct container liner shipping services servicing the ports in an economy. • <i>Number of alliances</i>: Number of direct container liner shipping alliances servicing the ports in an economy. • <i>Number of transshipments</i>: Number of times inbound containers in an economy are transshipped in their journey from the port of loading in the origin economy to the port of destination in the import economy.	• <i>Container export dwell time (days)</i>: Time spent by containers at the port of departure from arrival of full export container to completion of container loading on ship. • <i>Turnaround time at ports (days)</i>: Time container ships call at a port, excluding waiting time at anchorage outside the port vicinity. • <i>Time spent in transshipment (days)</i>: Time inbound containers in an economy spend in transshipment at intermediate port facilities during the shipping phase of its journey. • <i>Lead time to import from port of origin to destination economy (days)</i>: Time spent by containers between loading at the port of origin and discharge at the port of destination. • <i>Export supply chain initiation (days)</i>: Time between empty containers being sent to the exporter to be filled and full export containers being taken in charge by the shipping line at the port or inland facility. • <i>Import supply chain termination (days)</i>: Time between full containers being sent to the importer from the last logistics facility (port or inland) and empty containers being returned empty to the shipping line. • <i>Dwell time at port of arrival for container destined for a landlocked developing country (LLDC) (days)</i>: Average time spent by containers (destined for an LLDC) at the port of entry from arrival (ship unloading) to departure from the port. • <i>Corridor lead time from port exit to destination (LLDC) (days)</i>: Time spent by containers from the port of arrival to the clearance facility in the destination LLDC. • <i>Dwell time at destination (LLDC) (days)</i>: Import dwell time at the clearance facility in the destination LLDC. • <i>Reference dwell time at port of arrival (LLDC) (days)</i>: Import dwell time at the economy of the port of entry that is used most frequently for LLDC-destined cargo, calculated using only containers destined for the coastal economy.
Postal	• <i>Number of business-to-consumer postal partner countries</i>: Number of economies accessible through direct international postal connections for B2C shipments (letters and small packets up to 2 kilograms).	• <i>Business-to-consumer postal delivery time (days)</i>: Time spent by a B2C postal item in the importing economy from arrival at the economy's office of exchange (postal bureau) to the first attempted or final delivery.

Source: World Bank.

and lower logistics costs.²⁰ Three other maritime indicators gauge how economies could be affected by their dependence on third-country shipping hubs. These indicators are maritime lead time to import from port of arrival to the destination economy, time spent in transshipment, and number of transshipments for import containers. Two indicators that are derived from empty container movements—export supply chain initiation and import supply chain termination—reflect logistics efficiency among shippers, consignees, and trucking companies.

Two supplementary maritime time indicators capture additional aspects of performance at ports. Turnaround time (twenty-foot equivalent unit weighted and unweighted) reflects investment in port capacity and handling equipment, as well as the performance of terminal operations. Container export dwell time reflects several constraints that differ from those associated with import dwell time. Controls applied to export shipments are typically mild compared with those applied to imports, even in economies with long import delays. Formalities such as obtaining certificates of origin, which are processed upstream, are not automated in many economies and may delay containers. Furthermore, export dwell time includes buffer time, voluntary or mandated by the shipping lines (such as 48 hours in advance of ship departure). Factors such as export-import imbalances, distance to port, and frequency of shipping connections may influence private sector strategy and buffer time.

Traders in LLDCs face the challenge that their shipments usually travel long distances by land and cross land borders. To capture these challenges, the LPI 2.0 include four indicators specific to LLDCs: dwell time at port of arrival for a container destined for an LLDC, corridor lead time from port exit to destination, dwell time at destination (clearance facility in the LLDC), and reference dwell time at port of arrival. A comprehensive measure of container import time for LLDCs is the sum of the first three indicators. The fourth indicator measures import dwell time for containers destined for the coastal economy that is typically used as the transit economy for an LLDC. It is a benchmark for dwell

time at port of arrival for a container destined for an LLDC. Corridor lead time provides information on road or rail corridor performance.²¹

The supplementary postal logistics indicators are similar to the core indicators but focus on business-to-consumer (B2C) shipments, which are mostly e-commerce orders.²² B2C and B2B shipments typically face different customs procedures. Historically, B2C shipments (unlike B2B) benefited from *de minimis* exemptions.²³ But in recent years a growing number of economies have reconsidered these exemptions and increased scrutiny of B2C shipments, which has been associated with longer delays.

Data

Since the launch of the LPI in 2007, the rapid digitization of global supply chains has transformed the availability and granularity of logistics data. The 2025 LPI 2.0 leverage this shift by moving beyond perception-based surveys and drawing on large-scale shipment-level tracking data obtained under information-sharing agreements with global operators.²⁴ The World Bank collaborated with international carriers and data providers to assemble a multimodal dataset covering maritime shipping, air cargo, and postal services. This approach enables more objective, timely, and comparable indicators across economies.

The LPI 2.0 expand coverage relative to earlier editions, including many LLDCs and small island states that were previously excluded due to data gaps. Coverage remains limited for a small number of economies affected by embargoes or severe conflict, where global logistics operators have little or no presence. The dataset reflects actual logistics performance along international supply chains rather than national averages inferred from surveys.

Maritime indicators draw on several complementary sources. Container shipping connectivity is derived from MDS Transmodal's Containership Databank, which tracks container vessels and provides detailed information on service frequency and port rotations. Container dwell times, lead times, and supply chain initiation and termination indicators are calculated using shipment-level data from a major

global shipping line.²⁵ Port turnaround times are based on Automatic Identification System data from MarineTraffic, which records time-stamped vessel arrivals and departures within geofenced port areas, enriched with ship and port registry information.

Aviation and postal indicators complement the maritime data by capturing time and connectivity in high-value and small-parcel trade. Air cargo indicators are based on data from Cargo iQ, an International Air Transport

Association interest group, covering airport-to-airport shipments across more than 180 economies. Postal indicators rely on electronic tracking records from the Universal Postal Union, covering billions of international postal items annually and providing end-to-end time-stamps from posting to final delivery.

Together, these datasets form the empirical backbone of the LPI 2.0, enabling a more granular and evidence-based assessment of logistics performance across economies and modes.

What the LPI 2.0 reveals about logistics performance

This chapter presents high-level findings from the Logistics Performance Indicators 2.0 (LPI 2.0) with a focus on systematic patterns across regions and income groups. Data on all 21 indicators for 2023 and 2024 are published at <https://lpi.worldbank.org>, with options for comparisons across economies and over time.

Connectivity

On average, the Europe and Central Asia region and the North America region tend to have the highest connectivity across logistics modes, while the Latin America and the Caribbean region and the Sub-Saharan Africa region tend to have the lowest, depending on the mode considered (figure 2.1, panels a, b, c), although some developing economies perform strongly. This pattern mirrors broader income and geographic differences: high-income economies are generally more connected than low- and middle-income economies (figure 2.1, panels d, e, f). The connectivity indicators also reflect an economy's position in the global shipping network and its proximity to centers of economic activity.

Developing economies that have emerged as regional or global shipping hubs, such as Malaysia, Morocco, and Sri Lanka, have much higher connectivity than their neighbors with the same accessibility. For the same level of market accessibility, larger economies have more connections—for instance, Australia and New Zealand, which are among the economies with the lowest accessibility, have the highest connectivity at that level of accessibility (figure 2.2).

Connectivity increased during 2023–24 across most regions. Between 2023 and 2024 the number of direct liner shipping, aviation, and postal services connections increased across most income groups and regions, though some

economies in South Asia and in the Middle East and North Africa showed a small decline (see figure 2.1). These declines in maritime connectivity occurred largely because of the Red Sea crisis.

The existence of direct maritime connections matters because it removes significant delays associated with transshipments. Though about half of containers travel directly to their destination, the rest may be transshipped multiple times—up to eight in some cases (figure 2.3, panel a). Small island states are the most affected, with several averaging more than two transshipments in 2024: São Tomé and Príncipe (2.6), Anguilla (2.4), the Comoros (2.4), and St. Kitts and Nevis (2.2). Each transshipment adds to maritime lead times—roughly 18 extra days per transfer—underscoring the heavy burden of hub dependence for these economies (figure 2.3, panel b). But the hub and spoke structure of modern long-distance supply chains, where smaller freight loads are consolidated into larger ones at hubs using high-capacity equipment, has proven cost-effective for carriers while providing small and peripheral economies with access to global routes.

Time

Several low- and middle-income economies have moderate container import dwell times, similar to those in high-income economies, but many have exceptionally long ones (more than two weeks). Most of the economies with excessive dwell time are in Sub-Saharan Africa and in the Middle East and North Africa. Aviation import dwell time is typically only half of container import dwell time or less (figure 2.4). This difference reflects the smaller volume, smaller size, and higher value of air cargo compared with

Figure 2.1 Connectivity varies substantially across regions and income groups

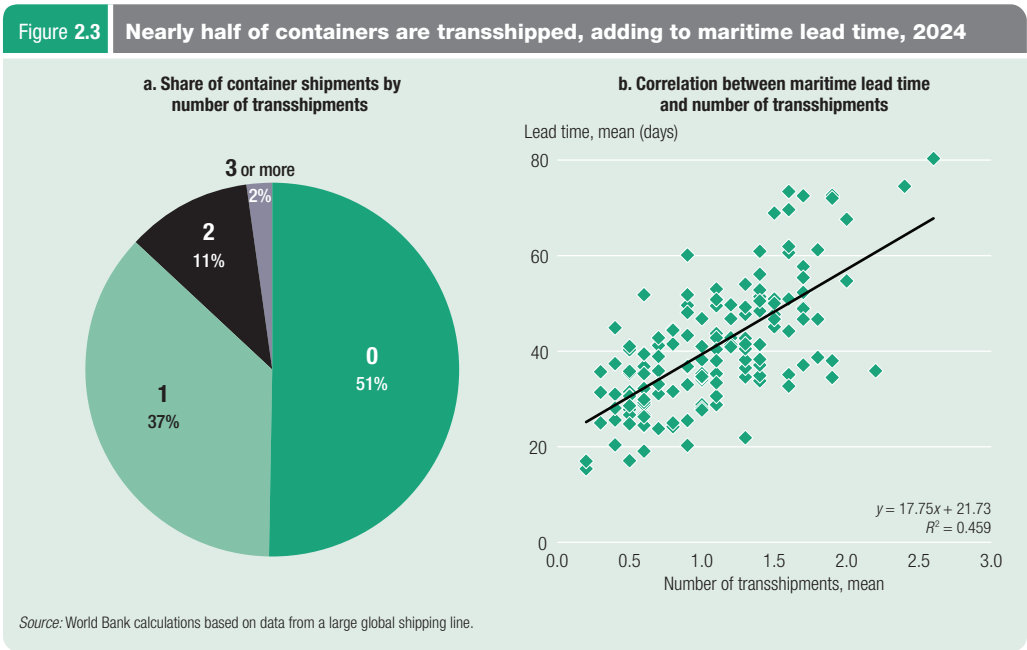
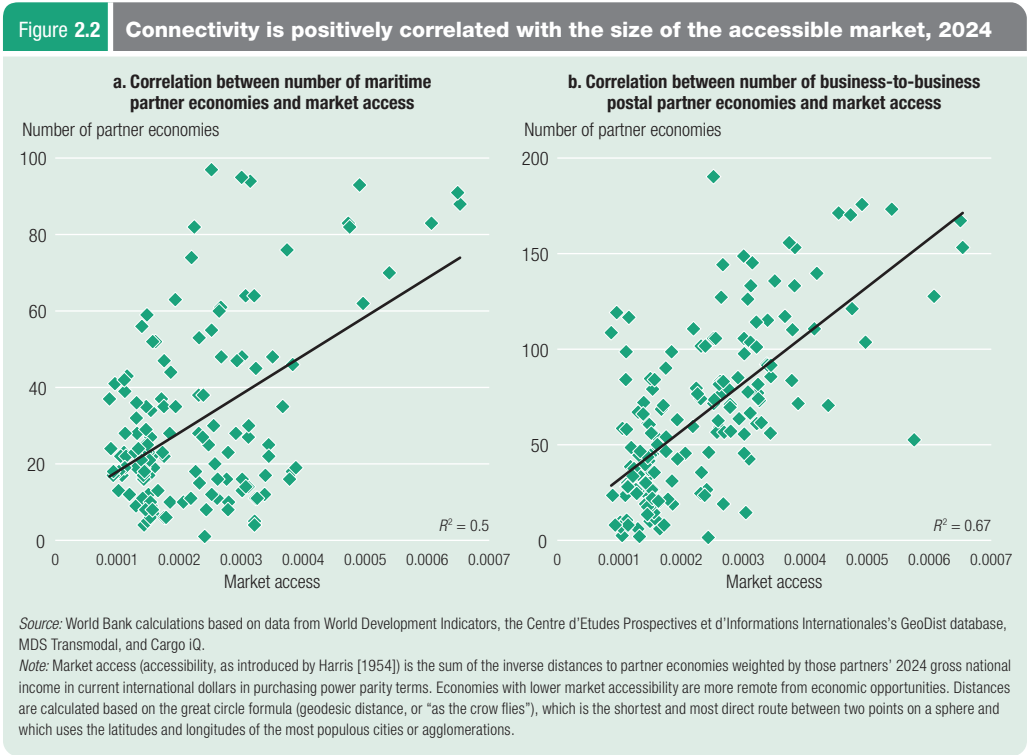
Source: World Bank calculations based on data from MDS Transmodal, Cargo IQ, and Universal Postal Union.

Note: Economy group averages are based on a balanced set of economies with data for both years. The numbers in parentheses are the number of economies included in the calculation of each group average.

maritime cargo. Air cargo operations are more tightly integrated with door-to-door logistics chains, and a share of shipments is cleared by customs before arrival, whereas maritime cargo is usually cleared after arrival. Underperforming economies overlap between the two indicators. The economies with the shortest dwell times are typically active air cargo hubs, such as Belgium, the Netherlands, and Singapore.

Long container import dwell times often reflect inefficient processes for clearing imported

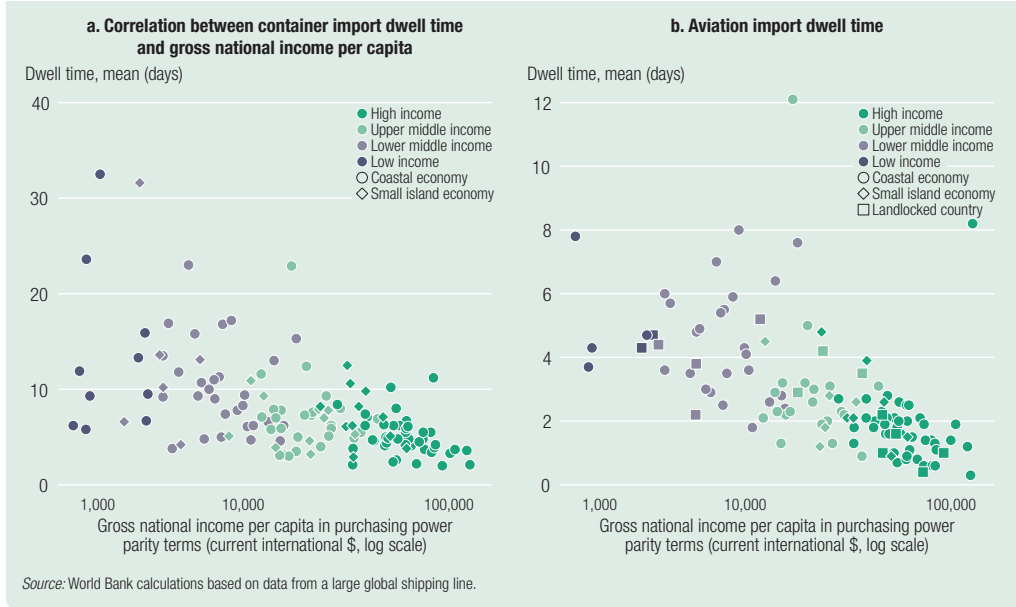
goods. Poor performance may stem from limited automation in customs clearance and port logistics—areas that port community systems can help improve.²⁶ Inadequate risk management and inefficient inspection practices by customs officials can also lengthen dwell times. Other requirements, including foreign exchange controls and inspections by agencies beyond customs, may further add to delays. Importer behavior can also contribute—for example, when importers use ports for storage.²⁷



Business-to-business (B2B) postal delivery time is highly dispersed across low- and lower-middle-income economies. While this pattern is common to other logistics modes, underperforming economies differ somewhat across modes. The economies with the longest container and aviation import dwell times are concentrated in Sub-Saharan Africa, whereas long B2B postal delivery

times are more common in Latin America and the Caribbean and in Sub-Saharan Africa. The differences between postal logistics performance and that of maritime containers and air cargo partly reflect the distinct facilities and processes involved in handling these different types of shipments. Two factors are especially important. First, geography and domestic distribution

Figure 2.4 Several middle-income economies have low import dwell times, but many have very high dwell times, 2024



matter: The postal indicator includes both the time spent at the postal bureau of entry and the time until delivery, which can be prolonged when domestic postal networks are not optimized, particularly in geographically large economies. Second, customs clearance practices play a key role. In some economies, including several in Latin America and the Caribbean, inbound shipment information is not electronically linked to customs systems, so clearance relies on paper documentation for each shipment, increasing delays.

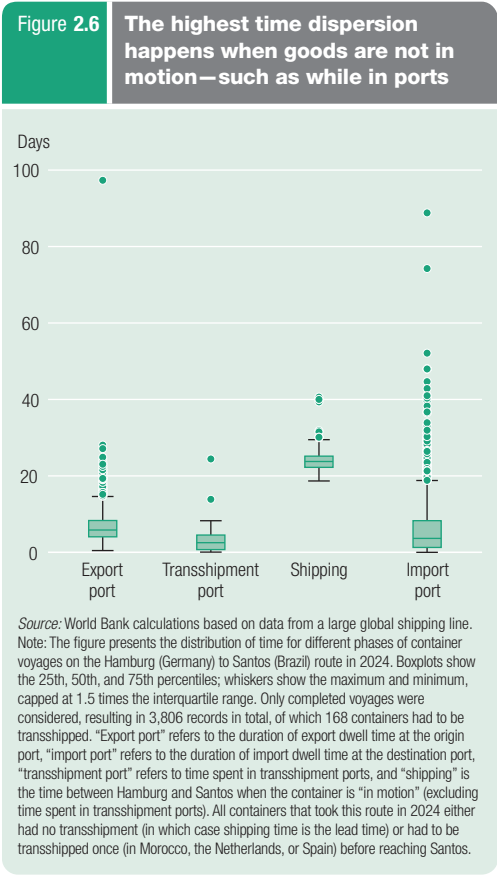
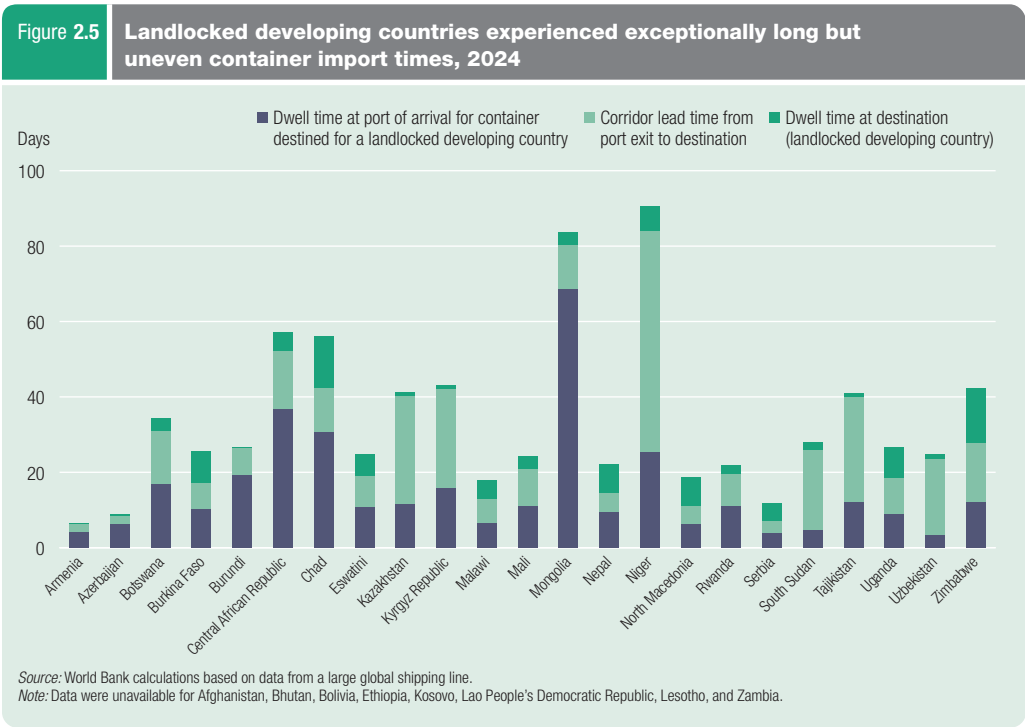
The special situation of landlocked developing countries

Landlocked developing countries (LLDCs) are among the most logistically constrained countries because their cargo needs to transit through coastal countries, crossing more than one border. As a result, the time to import a container is exceptionally long and uneven for LLDCs. Among the three components of time to import for LLDCs, corridor lead time and dwell time at port of arrival in coastal country tend to be the largest, followed by dwell time at destination (figure 2.5). In addition, dwell time at the port of arrival is longer for goods in transit than for goods destined for coastal countries.²⁸

The long import times are not explained by corridor distance but by the considerable time cargo spends idling at ports, at borders, and in checkpoints along transit corridors en route to the LLDCs—reinforcing the importance of pursuing corridor reforms. These reforms include tackling border crossing delays, improving exchange of information on transit vehicles and goods across borders, and adopting better transit procedures and guarantee mechanisms. Among the biggest technical challenges is cross-border interoperability of financial guarantees or bonds covering the customs value of goods in transit. The best practice systems are the Transports Internationaux Routiers system (managed by the International Road Union and used in Central Asia and the Balkans) and the Common Transit procedure (applicable to European Union and European Free Trade Association countries). Except for the Secretary of Economic Integration of Central America transit system, these systems have not been fully adopted by other regions.

Assessing supply chain reliability using the dispersion of delays

The highest dispersion of dwell, delivery, and lead times occurs when goods are not in motion:



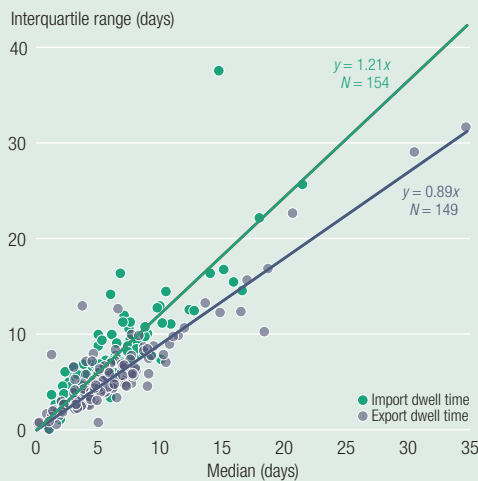
(figure 2.6).²⁹ Dispersion increases with median dwell times and is, on average, 36 percent higher for imports than for exports (figure 2.7). Dispersion is even higher for aviation supply chains.

In well-functioning ports with seamless clearance systems, processing an unloaded container starts immediately. In Le Havre (France) 10 percent of containers leave the port in 1 day or less, 50 percent do so in 3.8 days or less, and only 25 percent take more than 5.9 days (figure 2.8). By contrast, in many ports only a few containers are cleared during the first few days after arrival. In Apapa (Nigeria) only 0.2 percent of containers leave the port in 1 day or less, 50 percent do so in 13.4 days or less, and 25 percent take more than 22.1 days, and in Algiers (Algeria), only 0.1 percent leave the port in 1 day or less, 50 percent do so in 17.6 days, and 25 percent take more than 26.3 days.

Dispersion reflects the unpredictability of port clearance processes and port operations, with adverse implications for supply and value chains. Unreliable control processes, especially for clearing imported goods, are associated with the complexity of the processes, which often require collecting paper documentation and complying with complicated foreign exchange regulations. With export dwell time, factors include operational constraints, such as slot availability,

in ports, border controls, and inland checkpoints. The LPI 2.0 assess supply chain reliability at the economy level through the interquartile range of dwell, delivery, and lead times

Figure 2.7 Dispersion increases with the median dwell time and is higher for imports than for exports



Source: World Bank calculations based on data from a large global shipping line. Note: Data are for 2024. The lines are the fitted lines with 0 intercepts. N is the number of economies included. The formulas on the chart display the dependency of the interquartile range on median dwell time with the intercept of the fitted line forced to go through zero.

service frequency, and, in some economies, export clearances. Variations in shipper productivity and behavior can also affect the dispersion of export and import dwell time. The cost of unreliability can be substantial for firms' inventories and operational planning, especially in just-in-time supply chains and those handling perishable goods.

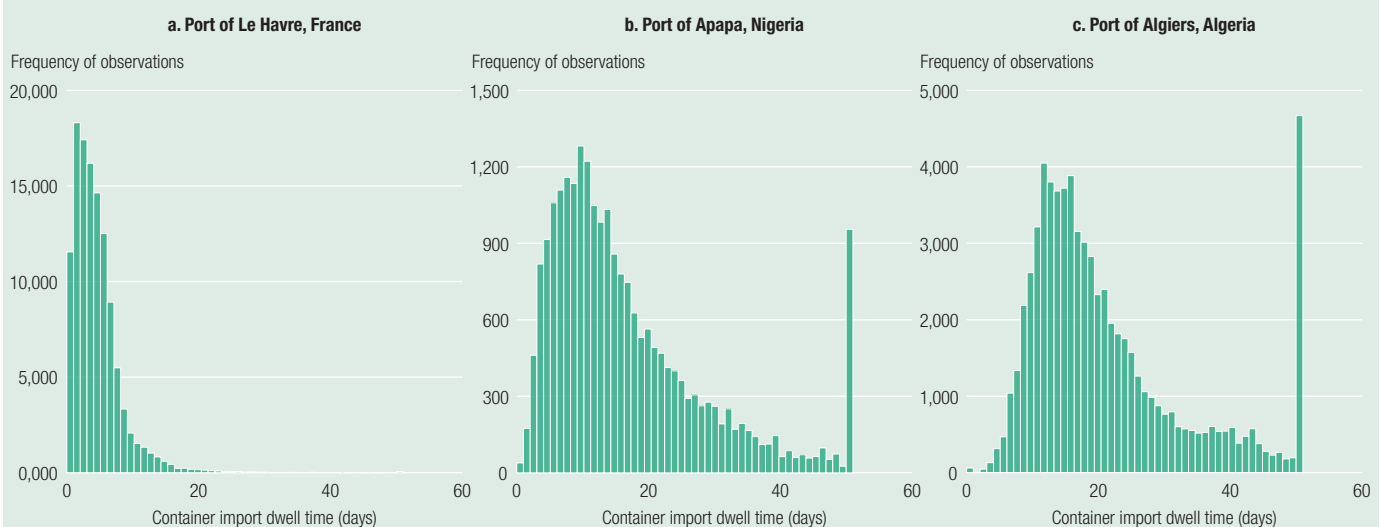
Tracking goods with information technology platforms, known as supply chain visibility, helps firms hedge some of the consequences of unreliability. Technology can help firms better anticipate the next steps in the supply chain. Examples of recent large-scale visibility system include the Freight Logistics Optimization Works (FLOW) initiative in the United States and the Unified Logistics Information Platform in India.³⁰

Additional insights from supplementary indicators

In some low- and middle-income economies container ships spend an excessive time at port (figure 2.9, panel a). Port turnaround time reflects port handling capacity and productivity and is more directly shaped by infrastructure capacity and terminal management practices than by trade facilitation measures. In most upper-middle- and high-income economies the average weighted turnaround time is less than two days, compared with two to six days in several low- and lower-middle-income economies. These differences point to substantial scope for improving container port performance in low- and lower-middle-income economies.³¹

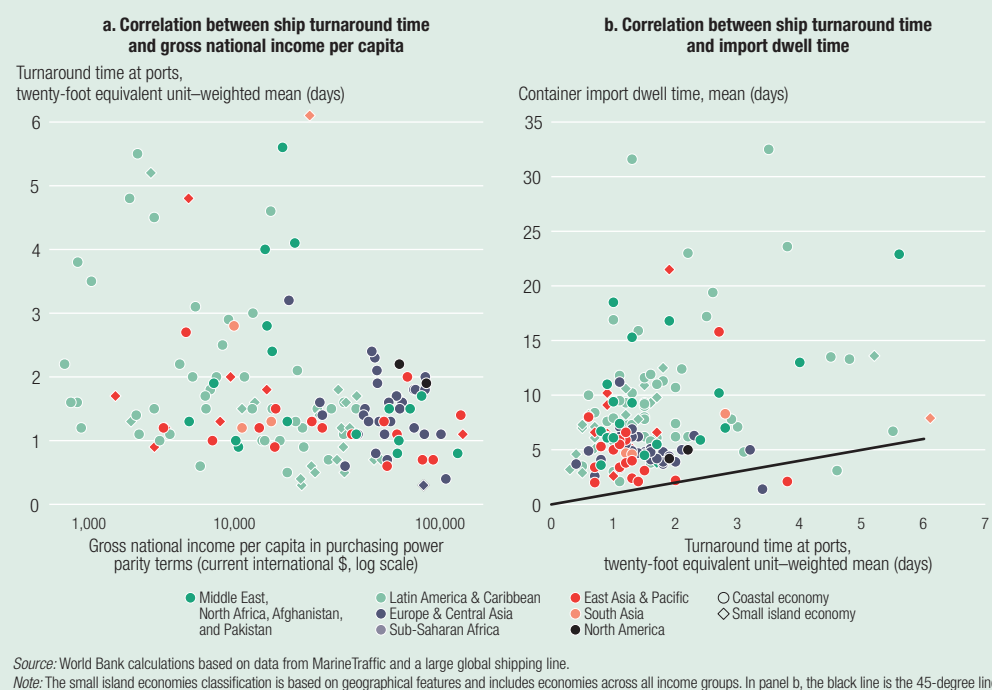
For most countries, however, the most painful frictions are in trade facilitation and

Figure 2.8 Import dwell time for the port of Le Havre (France), port of Apapa (Nigeria), and port of Algiers (Algeria), 2024



Source: World Bank calculations based on data from a large global shipping line. Note: Dwell times of 50 days and over are aggregated and shown at the "50" mark.

Figure 2.9 Average weighted ship turnaround time is shorter than import dwell time but still high in some economies, 2024



land-side logistics. This is evidenced by ship turnaround time typically being much shorter than container import dwell time (figure 2.9, panel b). Though dwell time is influenced by infrastructure, it captures primarily the performance of customs and other border procedures and therefore depends largely on trade facilitation policies.

Strong B2B and business-to-consumer (B2C) postal logistics performance is critical for e-commerce and SME participation. Although standard postal services experienced considerable market erosion in 2023 and 2024, they continue to play a substantial role in the last-mile delivery of cross-border e-commerce shipments (B2C), with close to 40 percent of these items being delivered by traditional postal services in 2024.³² Poor postal logistics performance tends to constrain small and medium enterprises (SMEs) from participating in e-commerce, with B2C SMEs usually finding it harder to scale up than B2B SMEs because B2C business entails a higher intensity of shipping.³³ Furthermore, B2C postal delivery times tend to be longer than B2B delivery times, primarily in lower-middle-income economies, pointing

to inconsistencies in postal logistics that penalize firms engaged in B2C e-commerce in these economies (figure 2.10).

Figure 2.10 Business-to-business postal delivery times tend to be longer than business-to-consumer postal delivery times, particularly in low- and middle-income economies, 2024



From indicators to policy interventions

Logistics services—such as freight transportation, warehousing, and brokerage—are delivered primarily by private firms. But their efficiency and the overall performance of supply chains are fundamentally shaped by public policy. Logistics operations depend on publicly funded infrastructure, must comply with border controls managed by state agencies, and operate within fiscal, environmental, safety, and competition regulations set by governments.

A strategic framework for logistics performance

Policy interventions to improve logistics performance, as measured by the LPI 2.0, should be implemented in a targeted, coherent manner across three strategic areas:

- Enhancing connectivity through strategic investment and liberalization.
- Reducing time and variability through trade facilitation and infrastructure.
- Strengthening the institutional and regulatory environment for logistics.

Together with reforms to reduce logistics costs—which are not directly measured by the LPI 2.0—these areas provide a practical policy agenda for governments seeking to improve logistics performance.

Area 1: Enhancing connectivity through strategic investment and liberalization

A country's connectivity to global production and distribution networks is determined by its physical geography, the quality and reach of its transport infrastructure, and the network strategies of international operators. While geography is fixed, public policy can greatly influence how well a country is connected to global markets.

Governments can act on three main fronts.

1.1 Liberalizing markets for transport and logistics services. Policies that open markets to competition and foreign participation can encourage international operators to expand their networks and increase the number of partner economies served. Examples include:

- Liberalizing maritime services and port operations to attract global shipping lines and terminal operators.
- Adopting “Open Skies” agreements in aviation to allow more carriers, routes, and frequencies.
- Removing restrictive regulations that limit entry or capacity in key logistics services (e.g., freight forwarding, express delivery).

These measures can increase service frequency, reduce costs, and improve the reliability of international connections.

1.2 Developing hubs, corridors, and logistics clusters. Governments can use infrastructure planning and spatial policies to position their economies as attractive nodes in regional and global networks. This involves:

- Investing in and upgrading gateway ports, airports, and dry ports that can serve as regional hubs.
- Developing multimodal transport corridors that connect these gateways to major production and consumption centers.
- Promoting logistics and industrial clusters around key hubs through zoning, serviced land, and streamlined permitting.

These policies can help transform a country's connectivity and establish it as a regional logistics and manufacturing hub.

1.3 Pursuing regional integration and shared connectivity solutions. For small, landlocked, or remote economies, national

policies alone may not be sufficient to achieve scale. Regional cooperation becomes essential. Key governments include:

- Harmonizing regulations and technical standards with partner economies to facilitate cross-border transport and logistics.
- Developing regional transit regimes and corridor management institutions to ensure seamless movement of goods.
- Coordinating investment in regional gateways and corridors so that smaller economies can “piggyback” on the trade flows of larger neighbors and global distribution centers.

By pursuing these policies, governments can systematically improve their connectivity performance, making their economies more attractive for trade and participation in global value chains.

Area 2: Reducing time and variability through trade facilitation and infrastructure

Time-related indicators—such as dwell time in ports and airports, ship turnaround time, and postal delivery time—are more responsive to policy than connectivity indicators are. Reducing delays and variability is critical for modern supply chains, especially those integrated into just-in-time production and e-commerce.

Governments can influence time and reliability through three main sets of policies.

2.1 Modernizing border management and trade facilitation. Border procedures are often a major source of delays and uncertainty. Key reforms include:

- Implementing risk-based management systems to focus inspections on high-risk consignments and expedite low-risk trade.
- Digitizing customs and border processes, including electronic submission of documents, prearrival processing, and electronic payment of duties and fees.
- Establishing coordinated border management, where customs and other border agencies (e.g., health, agriculture, security) align procedures and share information.

- For landlocked countries, designing and implementing efficient transit regimes, including transit guarantees, bonded transport, and streamlined procedures at border crossings.

These measures can greatly reduce import and export dwell times and improve the predictability of clearance processes.

2.2 Improving gateway efficiency and terminal operations. Even when border procedures are efficient, poor performance at ports, airports, and inland terminals can create bottlenecks. Government actions can include:

- Investing in infrastructure to address capacity constraints, such as berth extensions, yard expansion, and improved access roads and rail links.
- Promoting operational improvements, including port community systems, electronic appointment systems for truck arrivals, performance-based contracts with terminal operators, and efficient pricing of port storage.
- Using public–private partnerships (PPPs) to bring in private expertise and capital for terminal management, with clear performance targets for productivity and service quality.

These policies directly affect ship turnaround time, container dwell time, and the reliability of gateway operations.

2.3 Enhancing supply chain traceability and predictability. For firms participating in global value chains, predictability is often as important as speed. Governments can support more predictable logistics by:

- Addressing upstream causes of delays, such as complex banking procedures for foreign exchange, documentation requirements, or licensing processes that hold up shipments before they reach customs.
- Encouraging the use of digital tracking and tracing systems, including integration of customs, port, and logistics operator (including postal services) data.
- Supporting the development of express logistics and time-definite delivery services, which rely on streamlined customs procedures and integrated handling at gateways.

By focusing on both the direct and indirect sources of delay, governments can save time, reduce variability, and improve the reliability of supply chains.

Area 3: Strengthening the institutional and regulatory environment for logistics

Physical investments and trade facilitation reforms are most effective when supported by a sound institutional and regulatory environment. This area focuses on how governments organize, regulate, and coordinate the logistics sector. Governments can do so in three key policy areas.

3.1 Leveraging public–private partnerships and improving sector governance. PPPs, when well designed and embedded in a clear policy framework, can be powerful tools to improve logistics performance. Governments should consider:

- Using PPPs for port, airport, and inland terminal operations where private operators can bring management expertise, technology, and investment.
- Defining transparent concession frameworks with measurable performance indicators (e.g., crane productivity, truck turn-around time, dwell time) and mechanisms for monitoring and enforcement.
- Ensuring that PPPs are aligned with broader logistics and trade strategies rather than negotiated as isolated deals.

3.2 Strengthening competition, market structure, and supporting domestic logistics development. A competitive and professional logistics services market is essential for efficient supply chains. Policy priorities include:

- Removing unnecessary barriers to entry and anticompetitive regulations in trucking, freight forwarding, warehousing, and express delivery.
- Modernizing trucking regulations (e.g., axle load limits, vehicle standards, licensing) to balance safety and environmental objectives with efficiency.
- Opening markets to private and foreign providers where appropriate, while ensuring a

level playing field and effective competition oversight.

- Supporting the development of higher value-added logistics services (e.g., third-party logistics, contract logistics, cold chain) through clear regulations and investment-friendly policies.

These measures help improve domestic logistics performance and enable firms to access more sophisticated services, which in turn supports better connectivity and shorter delivery times.

3.3 Enhancing policy consistency, coordination, and implementation capacity. Connectivity and logistics performance improve gradually as operators adjust their networks and investment. Governments therefore need:

- Stable, predictable policy frameworks that give investors confidence to commit capital to infrastructure and logistics services.
- Institutional mechanisms—such as national logistics councils or trade facilitation committees—to coordinate across ministries (transport, trade, finance, customs, industry) and with the private sector.
- Clear national logistics or trade facilitation strategies with measurable targets linked to key performance indicators, such as those in the LPI 2.0, and systems to monitor progress and adjust policies as needed.
- Strengthened implementation capacity in customs, port authorities, transport agencies, and other relevant institutions, including training, change management, and adequate resourcing.

A coherent institutional and regulatory environment ensures that reforms under the first two areas are sustained and that their benefits are fully realized.

Using the LPI 2.0 to prioritize and tailor reforms

Logistics reform is not a one-size-fits-all endeavor. Economies differ in geography, economic structure, institutional capacity, and political economy. The LPI 2.0 provides a structured way to diagnose performance gaps in

connectivity and time and to link those gaps to specific policy levers.

By organizing interventions under three areas—enhancing connectivity, reducing time and variability, and strengthening institutions and regulations—governments can:

- Identify the root causes of logistics frictions.
- Prioritize the reforms with the highest expected impact on supply chain performance and trade outcomes.
- Sequence interventions in a way that reflects their capacity and socio-political context.

Ultimately, the value of the LPI 2.0 lies not only in measuring logistics performance but also in guiding governments toward concrete, evidence-based policy actions that improve connectivity, reduce time to trade, and support integration into regional and global value chains.

Notes

- 1 *Connecting to Compete 2023* featured a hybrid approach, presenting results from both the original LPI survey of logistics professionals and from supply chain tracking data. In 2025 the LPI 2.0 began using supply chain tracking data exclusively.
- 2 Arvis et al. 2013.
- 3 Hausman, Lee, and Subramanian 2013.
- 4 Korinek and Sourdin 2011.
- 5 Freund and Rocha 2010.
- 6 Hummels 2001. Manufactures exports refers to goods in categories 7 (machinery) and 8 (miscellaneous manufactures) of the Standard International Trade Classification. For all types of goods (not just manufactures), the decrease in likelihood is 1 percent.
- 7 Banomyong 2010.
- 8 World Bank 2020.
- 9 OECD 2015.
- 10 The World Bank's Global Supply Chain Stress Index measures the magnitude of container shipping disruptions affecting global supply chains.
- 11 Hummels, Lugovskyy, and Skiba 2009.
- 12 About 46 percent of global goods trade by value is moved in maritime containers (Notteboom, Pallis, and Rodrigue 2022), and 33 percent is moved by air (IATA 2025).
- 13 The LPI 2.0 do not cover road and rail logistics because no global source of data on those movements exists, but they cover container tracking on transport corridors, including some road transport (since the container tracking indicators indicate the quality of inland logistics services, including to LLDCs). While the first-best option would be end-to-end coverage of all shipments through all modes, the current reporting is the best available given the existing data. Future refinements of the LPI 2.0 could include data for other logistics modes.
- 14 For a detailed description of the methodology used to compute the indicators, see Arvis et al. (2026).
- 15 A liner shipping container service is a scheduled maritime transport service that moves standardized containers along fixed routes between designated ports on a regular and publicly advertised timetable. A direct service is a scheduled service between two economies that may include other stops in between but without a container needing transshipment.
- 16 Refers to direct flights.
- 17 For a given economy, postal connectivity is the average of two values: the number of unique origin partners from which postal items are received throughout a year and the number of partner economies to which postal items are shipped during the same period.
- 18 Removal refers to the process in which goods are removed from a warehouse or other location to make them available for the next step in the supply chain.
- 19 A maritime alliance (often called a liner shipping alliance) is a formal cooperative arrangement among independent container shipping lines to jointly operate services on major trade routes.
- 20 Hummels, Lugovskyy, and Skiba 2009.
- 21 It does not provide information on the performance of multimodal transfers en route, which are included in dwell time instead of lead time.
- 22 The LPI 2.0 differentiate between B2C and B2B flows because they are associated with different mail types. Parcels and express mail are attributed to B2B operations, whereas standard letter-post mail is attributed to B2C operations. Letter-post is used for the majority of e-commerce shipments. International e-commerce is interconnected with postal services: Universal Postal Union members are responsible for handling about two-thirds of letter-parcel (up to 2 kilograms) deliveries across borders. Universal Postal Union members also manage international express shipments, which account for 5–10 percent of international e-commerce volume, with the rest delivered by commercial operators (Beretzky et al. 2022).
- 23 The de minimis rule for imports allows low-value shipments to avoid tariffs and customs processing. For imports into the United States, this rule began being phased out in August 2025.
- 24 The agreements with the data partners stipulate that the World Bank may report summary statistics on the shipments but not individual shipment data.
- 25 The agreement with the data partner for container tracking precludes identifying the shipping line by name.
- 26 Ollivier, Sahu, and Saragiotis 2024. Port community systems are electronic platforms that connect the systems operated by the many organizations that constitute a seaport, airport, or inland port community (European Port Community Systems Association, n.d.).
- 27 Raballand et al. 2012.
- 28 See <https://lpi.worldbank.org/> for data on dwell time at port of arrival for goods destined for coastal economies.
- 29 The interquartile range measures the spread of the middle 50 percent of the distribution of delays. It is less sensitive to outliers than the standard deviation. Estimates of dispersion by economy and by indicator are available at <https://lpi.worldbank.org/>.
- 30 See <https://www.transportation.gov/freight-infrastructure-and-policy/flow> for details on the FLOW initiative. India's Unified Logistics Information Platform, launched in 2022 under the National E-Governance Plan, is a portal to make supply chain data generated by freight operators and government agencies accessible. It covers all logistics modes and encourages independent developers to propose visibility solutions for end users. The private sector has proposed more than 100 apps so far, highlighting the innovation potential of supply chain data.
- 31 For further discussion on container port performance, see World Bank (2024).
- 32 According to the Universal Postal Union.
- 33 Hendricks and Mwapwele 2024; UNCTAD 2018.

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What are the Logistics Performance Indicators (LPI 2.0)?

The LPI 2.0 is an interactive benchmarking tool created to help countries identify the challenges and opportunities they face in their performance on trade logistics and discover what they can do to improve their performance. Connecting to Compete 2025 introduces the LPI 2.0, a fundamental redesign of the World Bank's former Logistics Performance Index from a survey-based index toward a system grounded in shipment-level data. Drawing on large-scale tracking data from maritime, aviation, and postal operators—together accounting for over 80 percent of global goods trade—the LPI 2.0 provide standardized and comparable measures of supply chain speed, reliability, and connectivity across economies and over time. The framework comprises 21 indicators, including 6 core indicators capturing direct connectivity and import time performance in each mode and 15 supplementary indicators that shed light on maritime competition, transshipment dependence, postal logistics for e-commerce, and corridor performance for landlocked developing countries.

This is the eighth edition of Connecting to Compete, a report summarizing the findings from the new dataset for the Logistics Performance Indicators 2.0 (LPI 2.0). The 2025 LPI 2.0 draw on supply chain tracking data from the movers of international trade. This information is relevant for policymakers and decisionmakers in the private sector seeking to identify reform priorities for trade and logistics infrastructure. Data for 2023–24 reveal large variations in connectivity across regions, substantial time penalties associated with transshipment and border procedures, and high unpredictability concentrated at ports, transshipment hubs, and inland checkpoints—particularly for imports and for landlocked countries. By shifting the focus from perceptions to observed outcomes, the LPI 2.0 significantly enhance diagnostic precision and policy relevance, enabling governments to better prioritize reforms with the highest potential impact on supply chain connectivity, speed, reliability, and resilience.



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